

Optimal Demand Contracting Strategy Under Uncertainty and Its Implication for Advanced Pricing

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Abstract—Based on the current practice of charging peak demand, an expectation-oriented stochastic model is established for the demand contracting decision of electricity users. The resultant optimization problem is proved to be nonlinear but convex. Thus, the first-order optimality condition of the proposed model, which leads to an integral equation founded on the probability density function of the peak load, is used to derive the optimal contracting strategy for consumers. Simulation results support the convexity of the proposed model and the effectiveness of the proposed solution method under different demand charging rules and different consumption patterns of customers.

Index Terms—Demand side management, excessive demand, maximal demand, minimal charge, stochastic optimization.

NOMENCLATURE

Power Values

C	Transformer capacity of the customer.
$\tilde{D}$	Measured peak demand, which at the time of contracting is a random variable. $\tilde{D} \in [0, C]$.
$\underline{D}$	Minimal demand.
x	Contract demand.

Charging Parameters

c_s	Metering and services charge, whose unit is \$/connection.
p_0	Minimal demand price, whose unit is \$/kW.
p_1	Normal demand price, whose unit is \$/kW.
p_2	Excessive demand price, whose unit is \$/kW.

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k_1	Ratio threshold (to the contract demand), above which the normal demand price is enforced, below which the minimal demand price is used.
k_2	Ratio threshold (to the contract demand), above which the excessive demand price is enforced, below which the normal demand price is used.

Functions and Charging Components

$c(\tilde{D}, x)$	Demand charge.
$c_M(x)$	Minimal demand charge.
$c_N(\tilde{D}, x)$	Normal demand charge.
$c_E(\tilde{D}, x)$	Excessive demand charge.

Uncertainty Calibrations

$f_{\tilde{D}}(\bullet)$	Probability density function (PDF) of $\tilde{D}$.
$F_{\tilde{D}}(\bullet)$	Cumulative distribution function (CDF) of $\tilde{D}$.

I. INTRODUCTION

DEPENDING on how they use electricity, customers pay for different electric services, including but not limited to connection, energy, demand, and reactive power consumption.

Most industrial and commercial customers are required to pay for their peak demand (measured in kilowatts or kilovolt-amperes, abbreviated as kW or kVA), besides the energy they consume (measured in kilowatt-hours, abbreviated as kWh) [1]. Electric utilities charge them for the highest average demand measured in any 15 or 30 min during their billing period. Billing demand is based on consumers' measured maximal demand and their contract demand with utility companies as per the supply agreement. The specific charging rule varies between electric utilities.

Electricity users would like to minimize their electricity bills [2], [3]. Demand charge minimization are especially important for those large industrial customers whose electricity cost accounts for a significant part of their production cost, such as chemical fertilization, electric smelting, and aluminum electrolysis industries [4]–[8]. In mainland China, utilities allow large customers to adjust their contract demand

monthly [9]: if the peak demand does not exceed the contract demand, a fixed demand charge is levied; on the other hand, if the peak demand exceeds the contract demand, a penalty charge twice as the basic rate is levied.

Hence, choosing an extremely low contract demand will impose high demand charges, whereas choosing an excessively high contract demand may result in an unnecessary basic charge (BC). It is crucial for a rational consumer to adjust his/her contract demand to an optimal level in order to minimize his/her (expected) payment.

In Taiwan, an excess within 10% of the contract demand is charged at twice the rate of the contract demand, whereas the excess over 10% of the contract demand is charged at thrice the rate [10], and high-tension industrial customers can change their contract demand each month. Therefore, the problem of optimal contract decision has received significant attention [11]–[16].

In Southern Africa, the contract demand is determined by the notified maximum demand (NMD). Customers can temporarily or permanently increase/decrease their NMD, and their demand charge is based on the maximum of the measured demand and NMD [17]. Jemena Electricity Networks Ltd., of Victoria, Australia, also has a similar contract demand reset policy [18], which allows their customers to permanently/temporarily increase and permanently decrease their contract demand through request.

Based on our experience in Shanghai, the timely adjustment and notification of the contract demand is a valuable information source for utilities' load forecast and maintenance planning. Along with the development of advanced metering infrastructures and data analysis platform [19], it is expected that the contract demand can be adjusted more frequently and conveniently by a wider range of customers, so as to provide more accurate and complete information for smart grid operation and smart city functionalities.

However, customers have a limited idea of how much they will consume in the next billing period, especially those who has self-generations of wind and solar (these consumers are not uncommon among large industrial consumers who suffer demand charge, and will increase with the development of distributed generation and smart grids). Even for those customers with demand forecast capabilities, the forecast error still influences their payments.

Most existing studies [11]–[15] consider the problem of optimal demand contracting in a deterministic optimization framework, which is, in essence, a polynomial time linear programming [15]. It is emphasized in [16] that uncertain effects play an important role in the optimal energy management of a petrochemical plant, and formulates the problem into two layers. The outer layer is to determine the optimal contracting, the inner layer is to determine the optimal operation, and an interactive sampling-based Monte Carlo technique is used to simulate the uncertainties.

In this paper, an expectation-oriented analytical model is established to formulate the optimal demand contracting decision considering uncertainty. The resultant stochastic optimization problem is proved to be nonlinear but convex.

The nonlinearity of the problem depends on the stochastic characteristics of the peak demand of the next billing period. However, because the CDF is always increasing, or, equivalently, the PDF is always positive, the optimal demand contracting decision under uncertainty is a convex programming regardless of consumers' difference in their consumption pattern. Thus, the first-order optimality condition of the proposed model, which is an integral equation founded on the PDF of the peak load, is used to derive the optimal contracting strategy of the consumer. Based on this new framework, the proposed method shows desirable properties for demand contract decision: compared to deterministic methods, uncertainty is properly considered; compared with Monte Carlo simulation-based methods, the computation burden is significantly lower.

The stochastic optimization model for consumers' demand contracting strategy is proposed in Section II. The solution of the resultant stochastic program is strictly derived. The performance of the model and the solution strategy is further discussed in Section III by a numerical study. The convexity of the model and the existence of unique solutions are illustrated and validated under different conditions. Implications for advanced pricing practice of the regulatory authorities and utility companies have been drawn. Finally, Section IV concludes this paper.

II. METHODOLOGY

In this section, a general formulation for demand contracting decision is proposed. Demand charge of consumers is stochastic at the time of deciding their contract demand for the future billing periods; therefore, rational consumers' decision is to minimize their expected demand charge. The general formulation of the expected charge (EC) minimization model is thus provided. The resultant optimization problem is proved to be convex, and the optimal solution is strictly derived in general and under specific situations. Finally, the flow chart of a comprehensive solution procedure is provided for the optimal demand contracting decision under uncertainty.

A. General Model for Demand Charges

Though different from region to region, customers' demand charge, in general, can be categorized into three conditions: 1) the minimal charge; 2) the normal charge; and 3) the excessive charge.

The minimal charge is applicable when the peak demand is lower than the minimal billing demand, which depend on the demand of previous charging periods, a fixed kilowatt value, a certain percentage of transformer capacity, a certain percentage of the contract demand, or a combination of the above.

The normal charge is applicable when the peak demand is higher than the minimal billing demand but lower than the contract demand. The normal charge is consistent with the simple intuition that the more (demand) one uses, the more (demand charge) one pays; where under the minimal charge,

consumers pay the same no matter how much they use below the minimal billing demand.

The excessive charge is applicable when customers' measured demand is higher than their contract demand. The unit price for excessive demand is usually much more expensive than the normal rate (1.5~3.7 times of the normal rate in the regions studied in this paper).

Both the rate of normal charge and excessive charge can be multistep although single-step tariffs are also widely used [20]–[25]. Virginia, USA (II.A.2 customer) [26] and Gujarat, India (RATE HTP-I customer) [27] charges piecewise linearly, i.e., the normal rate is stepwise. In Virginia, if customers use more demand, they pay a lower unit fee; however, in Gujarat, if customers use more demand, they pay a higher unit fee. On the other hand, a stepwise excessive rate is adopted in Singapore, where customers pay a higher unit fee for more excessive demand [28].

Thus, a general formula of demand charge can be written as

$$\tilde{c} = c(\tilde{D}, x) = c_s + \begin{cases} c_M(x) + c_N(k_2 \cdot x, x) \\ \quad + c_E(\tilde{D}, x), & k_2 \cdot x < \tilde{D} \leq C \\ c_M(x) + c_N(\tilde{D}, x), & \max(k_1 \cdot x, \underline{D}) < \tilde{D} \leq k_2 \cdot x \\ c_M(x), & \tilde{D} \leq \max(k_1 \cdot x, \underline{D}). \end{cases} \quad (1)$$

Equation (1) shows that the demand charge ($c(\tilde{D}, x)$) is composed of four parts: 1) the service charge (c_s); 2) the minimal charge ($c_M(x)$); 3) the normal charge ($c_N(\tilde{D}, x)$); and 4) the excessive charge ($c_E(\tilde{D}, x)$).

The latter three parts can be formulated in general as

$$c_M(x) = p_0 \cdot \max(k_1 \cdot x, \underline{D}) \quad (2)$$

$$c_N(\tilde{D}, x) = p_1 \cdot \max(\tilde{D} - k_1 \cdot x, 0) + (p_{11} - p_1) \cdot \max(\tilde{D} - k_{11} \cdot x, 0) + (p_{12} - p_{11}) \cdot \max(\tilde{D} - k_{12} \cdot x, 0) + \cdots + (p_{1N} - p_{1(N-1)}) \cdot \max(\tilde{D} - k_{1N} \cdot x, 0) \quad (3)$$

$$c_E(\tilde{D}, x) = p_2 \cdot \max(\tilde{D} - k_2 \cdot x, 0) + (p_{21} - p_2) \cdot (\tilde{D} - k_{21} \cdot x, 0) + (p_{22} - p_{21}) \cdot \max(\tilde{D} - k_{22} \cdot x, 0) + \cdots + (p_{2E} - p_{2(E-1)}) \cdot \max(\tilde{D} - k_{2E} \cdot x, 0) \quad (4)$$

where $k_{11}, p_{11}; k_{12}, p_{12}; \dots; k_{1N}, p_{1N}$ denote the additional ratio thresholds and their corresponding prices when the normal charge is piecewise linear (rather than linear). $k_{21}, p_{21}; k_{22}, p_{22}; \dots; k_{2E}, p_{2E}$ denote the additional ratio thresholds and their corresponding prices when the excessive charge is piecewise linear (rather than linear).

Models (1)–(4) provide a framework to transform the practical demand charging rules into a mathematical form. Eleven real-world examples of models (1)–(4) are shown in Table I. One can notice that very different demand charging rules can be embodied by the proposed model.

TABLE I
PARAMETERS OF THE MODEL FOR DIFFERENT REGIONS

Region	$\underline{D}$	k_1	p_1	$k_{11}, p_{11};$ $k_{12}, p_{12};$...	k_2	p_2 $/p_1$	$k_{21}, p_{21};$ $k_{22}, p_{22};$...
South Dakota [23]	5kW	0	6.13 USD	N/A	N/A		
New York [22]	max(1kW, 0.5D-1)	0	8.32 USD	N/A	N/A		
Virginia [26]	50kW	0	2.668 USD	700, 2.130; 5000, 1.835	N/A		
Serbia [21]	N/A	1	486 RSD	N/A	1	2	N/A
Andhra Pradesh [24]	70kVA	0.8	350 INR	N/A	1	2	N/A
Karnataka [25]	50kW	0.75	180 INR	N/A	1	2	N/A
Gujarat [27]	100kV A	0	100 INR	500, 100; 1000, 140; 2500, 210	1	3.7	N/A
China [9]	0.4*C	0.95	39 CNY	N/A	1.05	2	N/A
Korea [20]	max(50kW, 0.5D-1)	1	5.71 KOR	N/A	1	1.5	N/A
Taiwan [10]	N/A	0	173.2 TWD	N/A	1	2	1.1, 5p1
Singapore [28]	850kW	1	6.05 SGD	N/A	1	1.5	1.2, 5p1; 2, 12p1

Model (1) clearly shows that the demand charge ($c(\tilde{D}, x)$) is composed of three parts: 1) the minimal charge ($c_M(x)$); 2) the normal charge ($c_N(\tilde{D}, x)$); and 3) the excessive charge ($c_E(\tilde{D}, x)$). Model (1) also reveals the mathematical forms of each part.

However, model (1) is a difficult starting point to conduct further analysis (one will be carried out in the following two subsections), because of the following.

- 1) Model (1) is general thus involves too many ($N_N + N_E + 7$) relevant parameters for an effective analysis.
- 2) The range definition of model (1) involves complicated logic judgment, it could be $k_1 \cdot x < \underline{D}$ or vice versa, thus one need first to judge the relationship between x and $\underline{D}$, then locate the range of $\tilde{D}$.

Such a double layer judgment would complicate the deduction carried out later.

Therefore, the following two assumptions are made.

- 1) $\underline{D}/k_1 < x < C/k_2$, though x could be any positive number, but only within this range is x effective.
- 2) $c_N(\tilde{D}, x)$ and $c_E(\tilde{D}, x)$ are both linear rather than piecewise linear, which makes the model compact.

Indeed, the conclusion obtained in the following two sections under this assumption still hold for the general piecewise linear model, because the EC of the linear and piecewise linear charge hold similar first-order and second-order derivative characteristics.

With the above two assumptions, a simpler formulation can be written as

$$\begin{aligned}\tilde{c} &= c(x, \tilde{D}) \\ &= c_s + \begin{cases} p_0 \cdot k_1 \cdot x + p_1 \cdot (\tilde{D} - k_1 \cdot x) \\ \quad + (p_2 - p_1) \cdot (\tilde{D} - k_2 \cdot x), & k_2 \cdot x < \tilde{D} \leq C \\ p_0 \cdot k_1 \cdot x + p_1 \cdot (\tilde{D} - k_1 \cdot x), & k_1 \cdot x < \tilde{D} \leq k_2 \cdot x \\ p_0 \cdot k_1 \cdot x, & \tilde{D} \leq k_1 \cdot x. \end{cases} \end{aligned} \quad (5)$$

One can notice that the parameter space of model (5) is much smaller (seven dimensions) than that of models (1)–(4), and the variable range judgments are all single layer, which makes the following analysis clear and easy to understand.

B. General Model for Expected Demand Charges

From model (5), one can notice that the demand charge, or consumers' demand payment, relates to C , $\underline{D}$, $\tilde{D}$, x , k_1 , k_2 , c_b , p_0 , p_1 , p_2 . From a consumer's perspective, k_1 , k_2 , c_b , p_0 , p_1 , p_2 are extraneous parameters determined by the regulatory authority or utility company; the transformer capacity C is almost unchangeable too at the time of decision. The measured demand of the billing period $\tilde{D}$ happens in the future of the decision time, thus is a stochastic variable affected by the natural and social environment, particularly weather and economic conditions. The only variable for which the consumer has full decision capability is the contract demand x . From the perspective of the contract demand x , model (5) can be rewritten as

$$\begin{aligned}\tilde{c} &= c(x, \tilde{D}) \\ &= c_s + \begin{cases} (p_0 \cdot k_1 - p_1 \cdot k_1 - p_2 \cdot k_2 \\ \quad + p_1 \cdot k_2) \cdot x + p_2 \cdot \tilde{D}, & x < \tilde{D}/k_2 \\ (p_0 \cdot k_1 - p_1 \cdot k_1) \cdot x + p_1 \cdot \tilde{D}, & \tilde{D}/k_1 \leq x < \tilde{D}/k_1 \\ p_0 \cdot k_1 \cdot x, & \tilde{D}/k_1 \leq x. \end{cases} \end{aligned} \quad (6)$$

In practice, usually $p_0 = p_1$, in this case model (6) can be simplified as

$$\begin{aligned}\tilde{c} &= c(x, \tilde{D}) \\ &= c_s + \begin{cases} (p_1 \cdot k_2 - p_2 \cdot k_2) \cdot x + p_2 \cdot \tilde{D}, & x < \tilde{D}/k_2 \\ p_1 \cdot \tilde{D}, & \tilde{D}/k_1 \leq x < \tilde{D}/k_1 \\ p_1 \cdot k_1 \cdot x, & \tilde{D}/k_1 \leq x. \end{cases} \end{aligned} \quad (7)$$

It is clear from (7) that the optimal decision is simply to contract a demand between $\tilde{D}/k_2$ and $\tilde{D}/k_1$, if the uncertainty of the future measured peak demand is excluded (in other words, the future peak demand is regarded as a fixed constant). This is consistent with the intuition and practical observations that customers want to contract closely to their measured peak demand.

However, customers' measured peak demand occurs in future, after they contracts or notifies a certain demand. Thus, the peak demand is random at the decision time. Therefore, the objective of minimizing demand charge is in fact to minimize

TABLE II
CHARACTERISTICS OF THE COMPONENTS IN MODEL (8)

Term	Range	Property	Monotone
EMC	≥ 0	Linear	+
ENC	≥ 0	Nonlinear	-
EEC	≥ 0	Nonlinear	-
BC	≥ 0	Constant	N/A

the expectation of demand charge, which can be derived from (5) as

$$\begin{aligned}\text{Min}_x \quad & E[\tilde{c}(x)] = c_s + p_0 \cdot k_1 \cdot x \\ & + \int_{k_1 \cdot x}^C p_1 \cdot (\tau - k_1 \cdot x) \cdot f_{\tilde{D}}(\tau) \cdot d\tau \\ & + \int_{k_2 \cdot x}^C (p_2 - p_1) \cdot (\tau - k_2 \cdot x) \cdot f_{\tilde{D}}(\tau) \cdot d\tau \quad (8) \\ \text{s.t.} \quad & \underline{D}/k_1 < x < C/k_2. \quad (9)\end{aligned}$$

Based on the PDF of peak demand, (8) reconciles three range definitions of (5) into three integral terms. In summary, the resultant EC includes four components.

- 1) $p_0 \cdot k_1 \cdot x$ indicates the expected minimal demand charge (EMC), which is linear to the contract demand x . Its slope $p_0 \cdot k_1$ is a constant decided by a product of the minimal demand price and the first ratio threshold.
- 2) $\int_{k_1 \cdot x}^C p_1 \cdot (\tau - k_1 \cdot x) \cdot f_{\tilde{D}}(\tau) \cdot d\tau$ indicates the expected normal demand charge (ENC), which is a nonlinear term. The derivative analysis shows that this term is monotonically decreasing.
- 3) $\int_{k_2 \cdot x}^C (p_2 - p_1) \cdot (\tau - k_2 \cdot x) \cdot f_{\tilde{D}}(\tau) \cdot d\tau$ indicates the expected excessive demand charge (EEC), which is a nonlinear term. The derivative analysis shows that this term is monotonically decreasing.
- 4) c_s still represents the BC of metering and services, which is the only fixed term.

Table II lists a summary of the mathematical characteristics of all the four components in model (8).

C. General Solution for the Optimal Contracting Strategy

In the previous subsection, the demand contracting decision is formulated as an expected cost minimization problem. The components of the objective function of this optimization model are analyzed in detail. In this subsection, the optimal solution of the model will be derived; the solution procedure turns out to be computationally attractive due to the particular properties of the proposed model.

Based on (8), the first-order derivative is written as

$$\begin{aligned}dE[\tilde{c}(x)]/dx &= p_0 \cdot k_1 - k_1 \cdot p_1 \cdot [1 - F_{\tilde{D}}(k_1 \cdot x)] \\ &\quad - k_2 \cdot (p_2 - p_1) \cdot [1 - F_{\tilde{D}}(k_2 \cdot x)]. \quad (10)\end{aligned}$$

The second-order derivative is

$$\begin{aligned}d^2E[\tilde{c}(x)]/dx^2 &= p_1 \cdot k_1^2 \cdot f_{\tilde{D}}(k_1 \cdot x) \\ &\quad + k_2^2 \cdot (p_2 - p_1) \cdot f_{\tilde{D}}(k_2 \cdot x) \quad (11) \\ &\because f_{\tilde{D}}(\bullet) \geq 0, k_1 > 0, k_2 > 0, p_2 \geq p_1 > 0, \therefore (11) \geq 0.\end{aligned}$$

A positive second-order derivative indicates that $E[\tilde{c}(x)]$ is globally convex.

On the other hand, (10) can be both positive and negative. Therefore, the first-order derivative is further classified. From (10), we have

$$\begin{aligned} dE[\tilde{c}(x)]/dx|_{x=C/k_2} &= p_0 \cdot k_1 - k_1 \cdot p_1 \\ &\quad \times [1 - F_{\tilde{D}}(k_1/k_2 \cdot C)] \\ &\stackrel{p_0 \equiv p_1}{=} k_1 \cdot p_1 \cdot F_{\tilde{D}}(k_1/k_2 \cdot C) \quad (12) \\ \because F_{\tilde{D}}(k_1/k_2 \cdot C) &> 0, k_1 > 0, p_1 > 0, \therefore (12) \geq 0 \\ dE[\tilde{c}(x)]/dx|_{x=\underline{D}/k_1} &= k_1 \cdot p_0 - k_1 \cdot p_1 \cdot [1 - F_{\tilde{D}}(\underline{D})] \\ &\quad - k_2 \cdot (p_2 - p_1) \\ &\quad \times [1 - F_{\tilde{D}}(k_2 \cdot \underline{D}/k_1)] \\ &\stackrel{p_0 \equiv p_1}{=} k_1 \cdot p_1 \cdot F_{\tilde{D}}(\underline{D}) - k_2 \cdot (p_2 - p_1) \\ &\quad \times [1 - F_{\tilde{D}}(k_2/k_1 \cdot \underline{D})]. \quad (13) \end{aligned}$$

There are two scenarios.

- 1) Equation (13) ≥ 0 , in this scenario $\because (11) \geq 0$, (12) ≥ 0 , and $dE[\tilde{c}(x)]/dx$ is continuous, $\therefore$ the global optimal is

$$x^* = \underline{D}/k_1. \quad (14)$$

- 2) Equation (13) < 0 , in this scenario $\because (12) \geq 0$ and $dE[\tilde{c}(x)]/dx$ is continuous, $\therefore \exists x^* \in [\underline{D}/k_1, C/k_2]$ holds $dE[\tilde{c}(x)]/dx|_{x^*} = 0$. Further because (11) = 0, this stationery point x^* is the global optimum, which satisfies

$$\begin{aligned} k_1 \cdot p_0 - k_1 \cdot p_1 \cdot [1 - F_{\tilde{D}}(k_1 \cdot x)] \\ - k_2 \cdot (p_2 - p_1) \cdot [1 - F_{\tilde{D}}(k_2 \cdot x)] = 0. \quad (15) \end{aligned}$$

Note that $F_{\tilde{D}}(\bullet)$ is an integral which generally cannot be expressed by elementary functions. Therefore, solving (15) implies solving an integral equation, which in general involves a numerical iterative method. In this paper, the Newton–Raphson method is adopted due to its simplicity and fast convergence, for this particular implementation, numerical integration is required in each step of the Newton–Raphson iteration.

In the conclusion, due to the reason that CDF is always increasing or, equivalently, PDF is always positive, the optimal demand contracting decision under uncertainty, though nonlinear, is a convex programming. Thus, the first-order optimality condition, which is an integral equation, is used to analyze and derive the optimal contracting strategy.

D. Analytical Solution Under Special Conditions

In the previous subsection, the general solution of the proposed model is discussed. In this subsection, two categories of real-world situations where the solution has simpler forms will be discussed.

- 1) The normal pricing section degenerates from a range to a point, which is true in Korea [20], Serbia [21], and Singapore [28]. Mathematically, this implies $k_1 = k_2$, then (8) degenerates to

$$\begin{aligned} E[\tilde{c}(x)] &= c_s + p_0 \cdot k_2 \cdot x \\ &\quad + \int_{k \cdot x}^C p_2 \cdot (\tau - k_2 \cdot x) \cdot f_{\tilde{D}}(\tau) \cdot d\tau. \quad (16) \end{aligned}$$

The first-order optimality condition degenerates to

$$dE[\tilde{c}(x)]/dx = p_0 \cdot k_2 - k_2 \cdot p_2 \cdot [1 - F_{\tilde{D}}(k_2 \cdot x)] = 0. \quad (17)$$

The stationary point is

$$x^* = F_{\tilde{D}}^{-1}(1 - p_0/p_2)/k_2. \quad (18)$$

Note that in (18), x^* has an explicit form, rather than implicitly bundled in an integral equation as in (15).

Let us further consider two special cases under this circumstance.

- a) *Case 1:* If $p_0/p_2 = 1$, then $x^* = 0$. The clear implication of this situation is: the optimal contract demand for the consumer shall be zero when the excessive price equals to the basic price, because no price difference will be enforced if the contract demand is exceeded.
 - b) *Case 2:* If $p_0/p_2 = 0.5$, then $x^* = F_{\tilde{D}}^{-1}(0.5)/k_2$, where $F_{\tilde{D}}^{-1}(0.5)$ indicates two-quantile or median. For a symmetric distribution, $F_{\tilde{D}}^{-1}(0.5) = E(\tilde{D})$. Further if $k_2 = 1$, $x^* = E(\tilde{D})$, that is, under this situation, a rational consumer will contract exactly at his/her expected demand to minimize the demand charge. The only real-world case falls into this case, i.e., simultaneously satisfy $p_0/p_2 = 0.5$ and $k_1 = k_2 = 1$, is the tariff in Serbia [21].
- 2) No excessive charge, which is true in New York [22] and South Dakota [23].

In the proposed model, this condition can be represented simply as $p_2 = p_1$, thus (8) degenerates to

$$\begin{aligned} E[\tilde{c}(x)] &= c_s + p_0 \cdot k_1 \cdot x \\ &\quad + \int_{k_1 \cdot x}^C p_1 \cdot (\tau - k_1 \cdot x) \cdot f_{\tilde{D}}(\tau) \cdot d\tau. \quad (19) \end{aligned}$$

The first-order optimality condition degenerates to

$$dE[\tilde{c}(x)]/dx = p_0 \cdot k_1 - k_1 \cdot p_1 \cdot [1 - F_{\tilde{D}}(k_1 \cdot x)] = 0. \quad (20)$$

Note that the only difference between (17) and (20) is that k_1 turns into k_2 . Thus, the unique stationary point of (19) is

$$x^* = F_{\tilde{D}}^{-1}(1 - p_0/p_1)/k_1. \quad (21)$$

Note that in (18) and (21), x^* has an explicit analytical form, unlike (15), where the optimal solution x^* is implicitly bundled in an integral equation or an optimization model.

E. Comprehensive Solution Procedure

In summary, rational consumers can contract a certain value of demand to minimize their expected demand charge in the future billing period(s). This optimal contract value can be under special conditions determined by direct analytical deduction, or in general determined by numerically solving an integral equation. The flow chart of a comprehensive procedure for optimal contracting decision under uncertainty is provided in Fig. 1.

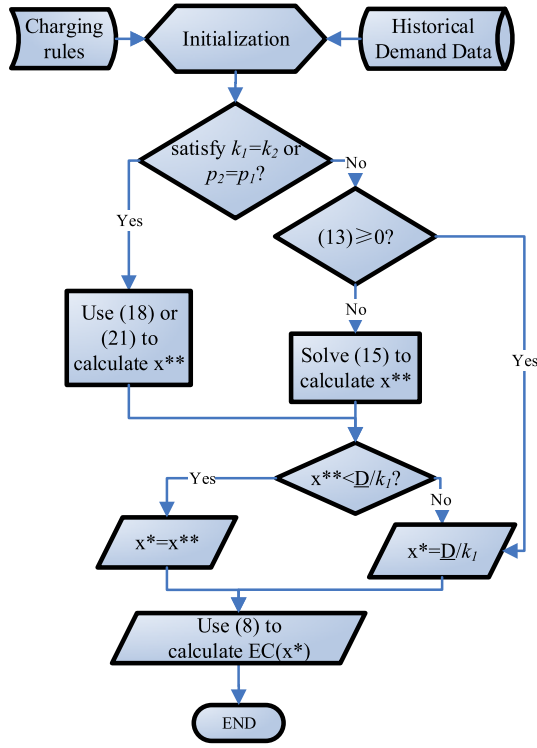


Fig. 1. Flow chart of the proposed algorithm.

TABLE III
BASE CASE

Charging Parameter	C	D	$p0$	$k1$	$p1$	$k2$	$p2$
Value	10	0	1	0.9	1	1.2	5

III. NUMERICAL CASES

In the previous section, a stochastic optimization model for consumers' demand contracting decision is proposed. The general solution of the proposed model is strictly derived. Components of the EC and their respective impact, as well as the properties of the solution under different conditions have been analyzed and discussed. In this section, the performance of the model and the solution strategy is further investigated through numerical examples. The convexity of the model and the existence of unique solutions are illustrated under different conditions. Implications for advanced pricing practice of the regulatory authorities and utility companies are also discussed.

For the purpose of clearness and comparableness, all the simulation results in this section are modified from a base case shown in Table III.

Note that the base case does not belong to any of the special situations listed in the previous section, in other words, the resultant optimality condition, i.e., (15), is an integral equation which does not have an analytical solution. A Newton–Rapson-based numerical method is adopted in this section to calculate the optimal contract value. For comparison, direct calculation based on (8) is used to generate the overall shape of the EC curve, so that to examine the effectiveness of the proposed method (optimal contract value on

the EC curve or not) and demonstrate the convexity under different situations.

A. Influence of the Uncertainty

As shown in Section II-C, uncertainty of the peak demand, which can be calibrated by its PDF or CDF, significantly influences the optimal demand contracting strategy of a rational consumer.

In practice, PDF/CDF of the future peak demand can be drawn in two ways: for an old consumer with sufficient historical data of its own monthly peak demand data, the most direct way is distribution fitting, i.e., the fitting of a probability distribution to histogram of historical demand data concerning the repeated measurement of an occurrence; more sophisticated methods include probabilistic forecast, which still utilizes historical demand data, but also integrates calendar effects, weather conditions, and/or economic variables in advanced manners [29]–[32]. For a new consumer without enough historical data of its own consumption, the PDF/CDF of those consumers of a similar type can be very helpful to form the PDF/CDF of its own future peak demand.

The numerical study here include the following typical distributions, which are widely used for representing electricity use of a single consumer, namely lognormal [33], [34], Weibull [35], [36], gamma [34], [37], and beta [38]–[40] distributions.

The PDF of a Weibull random variable x is defined as

$$f_W(x) = \begin{cases} k/\lambda \cdot (x/\lambda)^{k-1} e^{-(x/\lambda)^k}, & x > 0 \\ 0, & x \leq 0 \end{cases} \quad (22)$$

where $k > 0$ is the shape parameter and $\lambda > 0$ is the scale parameter. The integral of (22), i.e., the CDF of $x \sim \text{Weibull}(\lambda, k)$ is

$$F_W(x) = \begin{cases} 1 - e^{-(x/\lambda)^k}, & x > 0 \\ 0, & x \leq 0 \end{cases} \quad (23)$$

Exponential distribution and Rayleigh distribution are two special cases of Weibull distribution where $k = 1$ and $k = 2$, respectively.

The PDF of a random variable x which follows beta distribution can be defined as:

$$f_B(x) = \begin{cases} \frac{\Gamma(\alpha + \beta)}{\Gamma(\alpha) \cdot \Gamma(\beta) \cdot C^{\alpha + \beta - 1}} x^{\alpha - 1} (C - x)^{\beta - 1}, & 0 < x < C \\ 0, & x \leq 0 \cup x \geq C \end{cases} \quad (24)$$

where $\alpha > 0$ and $\beta > 0$. The CDF of $x \sim \text{Beta}(\alpha, \beta)$ is

$$F_B(x) = \begin{cases} 0, & x \leq 0 \\ \int_0^{x/C} \tau^{\alpha - 1} (1 - \tau)^{\beta - 1} / \int_0^1 \tau^{\alpha - 1} (1 - \tau)^{\beta - 1}, & 0 < x < C \\ 1, & x \geq C. \end{cases} \quad (25)$$

The PDF of a lognormal random variable x is defined as

$$f_L(x) = \begin{cases} \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{\ln x - \mu}{\sigma} \right)^2}, & x > 0 \\ 0, & x \leq 0 \end{cases} \quad (26)$$

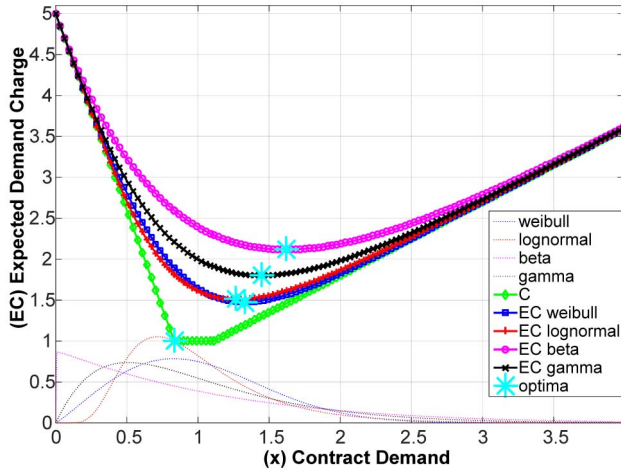


Fig. 2. Impact of uncertainty, base case.

where $\mu > 0$ and $\sigma > 0$. The CDF of $x \sim \text{Lognormal}(\mu, \sigma)$ is

$$F_L(x) = \frac{1}{2} \cdot \text{erfc}\left(-\frac{\ln x - \mu}{\sigma \sqrt{2}}\right) = \Phi\left(\frac{\ln x - \mu}{\sigma}\right) \quad (27)$$

where $\text{erfc}(\bullet)$ denotes the complementary error function and $\Phi(\bullet)$ is the CDF of the standard normal distribution.

The PDF of a gamma random variable x is defined as

$$f_G(x) = \frac{x^{k-1} e^{-x/\theta}}{\theta^k \Gamma(k)} \quad (28)$$

where $\Gamma(\bullet)$ denotes the gamma function, $k > 0$ and $\theta > 0$ are the shape and scale parameters, respectively.

The CDF of $x \sim \text{Gamma}(k, \theta)$ is

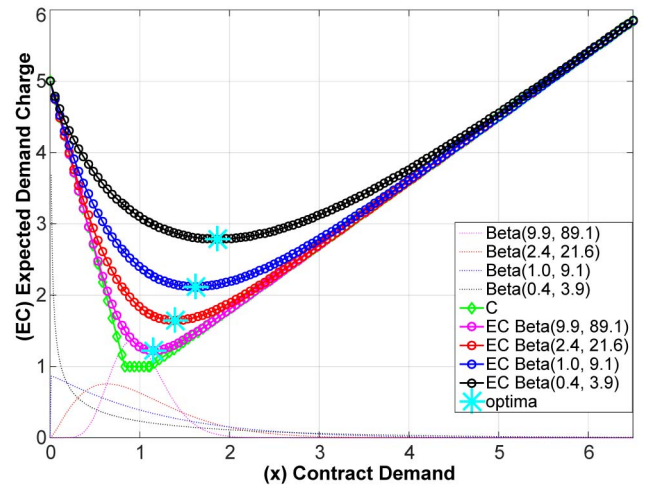
$$F_G(x) = \frac{\gamma(k, x/\theta)}{\Gamma(k)} \quad (29)$$

where $\gamma(\bullet, \bullet)$ denotes the lower incomplete gamma function.

The simulation results are shown in Fig. 2: the magenta circled curve, the red plus-signed curve, the blue squared curve, black crossed curve demonstrate the expected demand charge when the demand follows $\text{lognormal}(-0.11, 0.47)$, $\text{Weibull}(1.13, 2.1)$, $\text{gamma}(2.04, 0.49)$, and $\text{beta}(1.01, 9.1)$ distributions, respectively. The parameters are calibrated to let these different distribution hold the same expectation as one, i.e., $E(\tilde{D}) = 1$, and the first two hold the same standard deviation, while the latter two hold steadily larger deviations, i.e., $\sigma(\tilde{D}) = [0.5, 0.5, 0.7, 0.9]$. The green diamond curve show the expected cost if future demand is certain as one, i.e., $\tilde{D} = 1$, as a benchmark for comparison.

The following can be noticed in Fig. 2.

- 1) Different uncertainty/distribution may result in different shapes of the EC versus x curve, these curves may even cross each other; however, the curves are always convex, which is consistent with the results of analytical deduction in Section II-C, resulting in unique contracting strategies. (In contrast, the certainty case result in a three section piecewise linear curve and multiple optimal solutions $x^* = [D/k_1, D/k_2] = [0.83, 1.11]$.)
- 2) When uncertainty exists, the EC curve will be lifted. The cost curve of the certainty case is always lower (along

Fig. 3. Base case, beta distribution, $\sigma = [0.3 \ 0.6 \ 0.9 \ 1.3]$ illustrated in [magenta, red, blue, and black] lines, respectively.

all the contract demand values) than all the uncertainty cases, no matter what the specific probability distribution of the uncertainty is. The cost gap between the certainty case and the uncertainty cases, or uncertainty uplift, however, varies between different contract demands.

- 3) The optimal contracting value under uncertainty is always larger than that under certainty, denoting the inclination of avoiding EEC, at the stake of increasing EMC and ENC.

The latter two observations are further demonstrated in Fig. 3, where the standard deviation increases from 0.3 to 1.3 and the expectation still hold the same as 1. For clearness of illustration, only the results of beta distribution are retained. The resulting distributions are $\text{Beta}(9.9, 89.1)$, $\text{Beta}(2.4, 21.6)$, $\text{Beta}(1.0, 9.1)$, and $\text{Beta}(0.4, 3.9)$, shown in magenta, red, blue, and black circled curves, respectively. It can be noticed that the EC curve is very close to the cost curve of certainty when the standard deviation, a measure of uncertainty, is small. As the standard deviation increases, EC curve will be lifted in the middle. Accordingly, the optimal contract value increases from 1.17 up to 1.86, and the minimal EC increase from 1.15 up to 2.79.

B. Influence of the Charging Parameters

In the base case, it is assumed that $p_0 = p_1$, which is consistent with all the real-world cases studied in this paper. But for theoretical completeness, it is also worth checking the results when $p_0 \neq p_1$. As Fig. 4 ($p_0 = 1 < p_1 = 1.5$) and Fig. 5 ($p_0 = 1.5 > p_1 = 1$) show, the cost of certainty case is still a three section piecewise linear curve, but the middle section becomes increasing or decreasing (rather than flat), and unique optimal contract values exist at $x^* = D/k_2 = 1.11$ and $x^* = D/k_1 = 0.83$, respectively.

Fig. 6 shows another special situation when $k_1 = k_2 = 1$, under this condition, the middle section degenerates from a range to a point, but the convexity of the EC curves still holds.

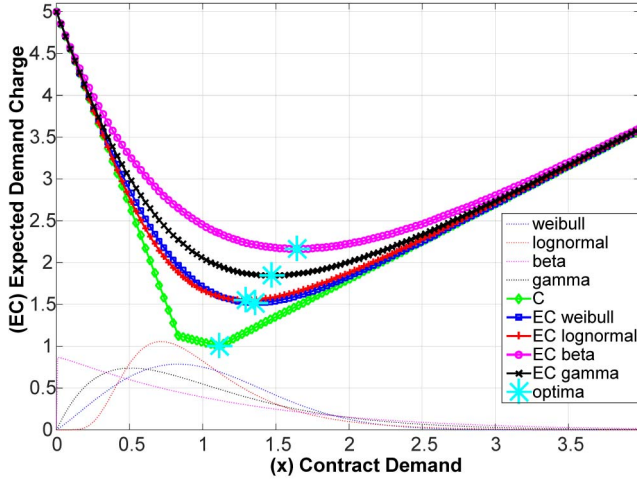
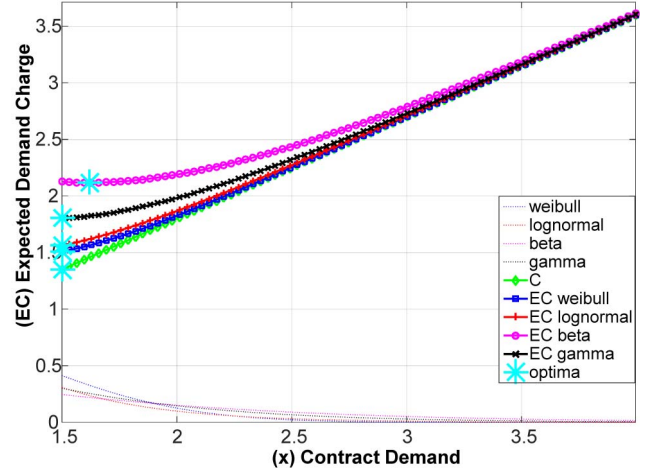
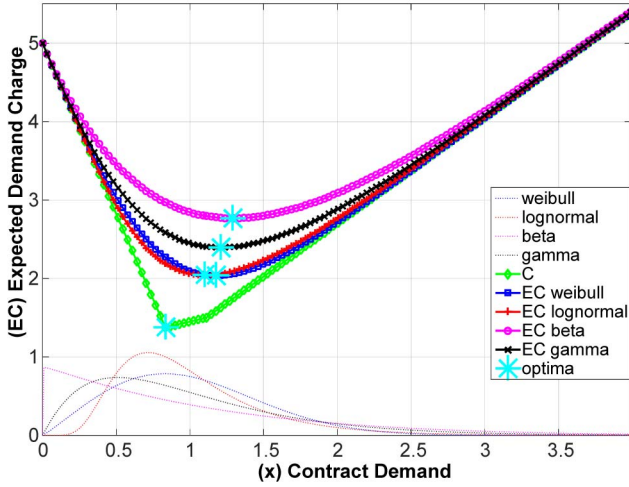
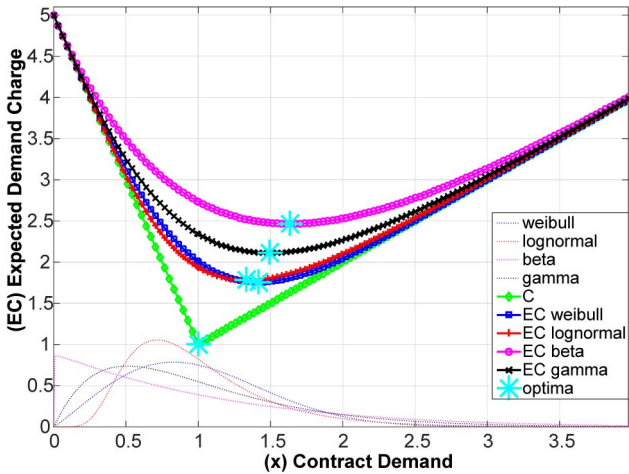
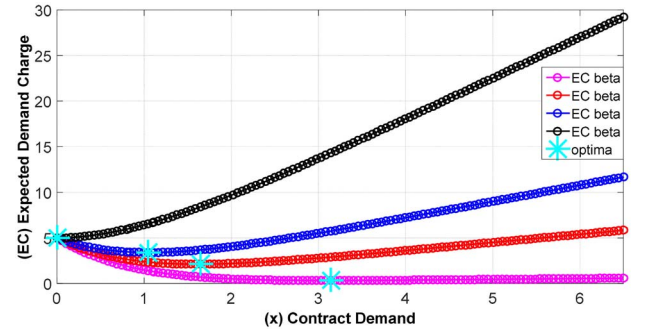
Fig. 4. Base case, $p_0 = 1 < p_1 = 1.5$.Fig. 7. Base case, $k_0 = 0.15$.Fig. 5. Base case, $p_0 = 1.5 > p_1 = 1$.Fig. 6. Base case, $k_1 = k_2 = 1$.

Fig. 7 shows the relationship between the EC and the contract demand when the minimal demand shifts from 0 to 1.5. One can notice that this modification changes

Fig. 8. Base case, Beta(1.01, 9.1), $p_0 = p_1 = [0.1 \ 1 \ 2 \ 5]$ illustrated in [magenta, red, blue, and black] lines, respectively.

neither the location nor the shape of the EC curve. The only impact is the effective range of the contract demand. Therefore, the optimal contract value of the Beta(1.01, 9.1) consumption stays the same, which is a case of the scenario 2) of section II.C, while the optimal contract value of lognormal(−0.11, 0.47), Weibull(1.13, 2.1), and gamma(2.04, 0.49) moves to $x^* = \underline{D} = 1.5$, corresponding to the scenario 1) of Section II-C.

Fig. 8 shows the relationship between the EC and the contract demand when normal demand price p_1 varies. One can notice that the left end ($x = 0$, EC = 1.1) stays the same, but the right part of the EC curve varies. To be exact, the right asymptote still performs a linear curve but with greater slopes as p_1 increases. With this effect, the optimal contract value decreases and the minimal EC value increases as p_1 increases.

Fig. 9 shows the relationship between the EC and the contract demand when p_2 varies. One can notice that the right asymptote of the EC curve stays the same but the left end of the EC curve increases. With this effect, both optimal contract value and minimal EC value increases as p_2 increases.

Fig. 10 shows the relationship between the EC and the contract demand when k_1 varies. One can notice that the left end stays the same, but the right asymptote still performs a linear curve but with greater slopes as p_1 increases. With this effect, the optimal contract value decreases and the minimal EC value increases as k_1 increases.

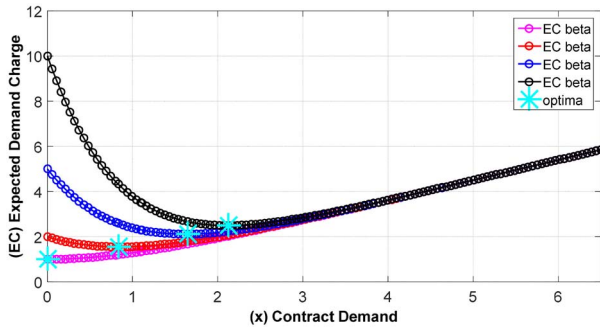


Fig. 9. Base case, Beta(1.01, 9.1), $p_2 = [1.2510]$ illustrated in [magenta, red, blue, and black] lines, respectively.

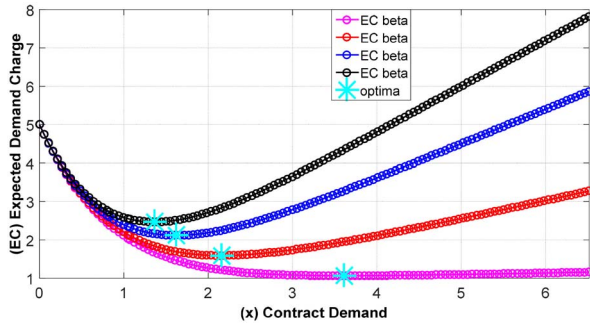


Fig. 10. Base case, Beta(1.01, 9.1), $k_1 = [0.10.50.91.2]$ illustrated in [magenta, red, blue, and black] lines, respectively.

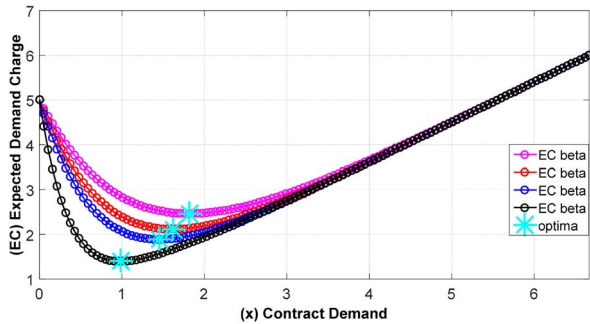


Fig. 11. Base case, Beta(1.01, 9.1), $k_2 = [0.91.21.53]$ illustrated in [magenta, red, blue, and black] lines, respectively.

Fig. 11 shows the relationship between the EC and the contract demand when k_2 varies. One can notice that the left end and the right asymptote stay the same, but the middle part of the curve varies. To be exact, the middle part of the charge curve still performs a convex nonlinear curve but with a larger curvature as k_2 increases. With this effect, both optimal contract value and minimal EC value decreases as k_2 increases.

Let us summarize this section in two aspects.

From practical aspect, regulatory authority can modify the structure, prices, and thresholds of the demand charging rule so as to influence rational consumers' demand contracting strategy and expected demand charge. The former is a useful information for forecasting the consumers' peak load, while the latter, can be used to balance the supplying cost of electricity utilities, especially the capacity investment cost. For the recovery of capacity cost, adjusting demand prices

(p_0, p_1, p_2) is a classical way; the results in this section indicate that similar effect can be achieved through purely adjusting the structure or the threshold parameters (k_1, k_2) .

From theoretical aspect, though the shape, curvature, left, and right end points of the EC curve will be influenced by the demand charging rules, the convexity of the EC curve always holds, therefore, the existence and uniqueness of the optimal contract value are always ensured in the proposed model.

IV. CONCLUSION

This paper proposes a stochastic programming model and its corresponding solution for electricity consumers' optimal demand contracting decision under uncertainty. The initial results and findings, as well as their implication for charging practice are thoroughly discussed. Future work is to develop a real-time pricing mechanism for excessive demand consumption, so that to further improve the interaction between consumers and distribution network operators.

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REFERENCES

- [1] J. Wang, C. Bloyd, Z. Hu, and Z. Tan, "Demand response in China," *Energy*, vol. 35, no. 4, pp. 1592–1597, Apr. 2010.
- [2] A. Fakhrazari, H. Vakilzadian, and F. Choobineh, "Optimal energy scheduling for a smart entity," *IEEE Trans. Smart Grid*, vol. 5, no. 6, pp. 2919–2928, Nov. 2014.
- [3] A. Staden, J. Zhang, and X. Xia, "A model predictive control strategy for load shifting in a water pumping scheme with maximum demand charges," *Appl. Energy*, vol. 88, no. 12, pp. 4785–4794, Dec. 2011.
- [4] M. Paulus and F. Borggrefe, "The potential of demand-side management in energy-intensive industries for electricity markets in Germany," *Appl. Energy*, vol. 88, no. 2, pp. 432–441, Feb. 2011.
- [5] W. Kong, T. Chai, J. Ding, and S. Yang, "Multifurnace optimization in electric smelting plants by load scheduling and control," *IEEE Trans. Autom. Sci. Eng.*, vol. 11, no. 3, pp. 850–862, Jul. 2014.
- [6] S. Ashok and R. Banerjee, "An optimization mode for industrial load management," *IEEE Trans. Power Syst.*, vol. 16, no. 4, pp. 879–884, Nov. 2001.
- [7] S. Ashok, "Peak-load management in steel plants," *Appl. Energy*, vol. 83, no. 5, pp. 413–424, May 2006.
- [8] C. A. Babu and S. Ashok, "Peak load management in electrolytic process industries," *IEEE Trans. Power Syst.*, vol. 23, no. 2, pp. 399–405, May 2008.
- [9] Ministry of Power Industry of the People's Republic of China, Decree No. 8, *Power Supply Business Rules (Clause no. 85)*. [Online]. Available: http://www.nea.gov.cn/2012-01/04/c_131262676.htm
- [10] Rate Schedules, Taiwan Power Comp., Taipei, Taiwan, 2014. [Online]. Available: <http://info.taipower.com.tw/indexE.htm>
- [11] H. Yang and P. Peng, "Improved Taguchi method based contract capacity optimization for industrial consumer with self-owned generating units," *Energy Convers. Manage.*, vol. 53, no. 1, pp. 282–290, 2012.
- [12] T. Lee and N. Chen, "Determination of optimal contract capacities and optimal sizes of battery energy storage systems for time-of-use rates industrial customers," *IEEE Trans. Energy Convers.*, vol. 10, no. 3, pp. 562–568, Sep. 1995.
- [13] M. Tsay, W. Lin, and J. Lee, "Optimal contracts decision of industrial customers," *Int. J. Elect. Power Energy Syst.*, vol. 23, no. 8, pp. 795–803, 2001.
- [14] T. Lee and C. Chen, "Iteration particle swarm optimization for contract capacities selection of time-of-use rates industrial customers," *Energy Convers. Manage.*, vol. 48, no. 4, pp. 1120–1131, 2007.
- [15] C. Chen and C. Liao, "A linear programming approach to the electricity contract capacity problem," *Appl. Math. Model.*, vol. 35, no. 8, pp. 4077–4082, 2011.

- [16] T. Wu, S. Shieh, S. Jang, and C. Liu, "Optimal energy management integration for a petrochemical plant under considerations of uncertain power supplies," *IEEE Trans. Power Syst.*, vol. 20, no. 3, pp. 1431–1439, Aug. 2005.
- [17] Eskom. *Notified Maximum Demand (NMD) Rules*. [Online]. Available: http://www.eskom.co.za/CustomerCare/TariffsAndCharges/Pages/Notified_Maximum_Demand_NMD_Rules.aspx
- [18] Jemena Electricity Networks (VIC) Ltd, *Policy for Resetting Contract Demand*. [Online]. Available: <http://jemena.com.au/>
- [19] *Three-Year Action Plan of Big Data Research and Development*, Shanghai Sci. Technol. Comm., Shanghai, China, 2013–2015. [Online]. Available: <http://www.stcsm.gov.cn/gk/ghjh/333008.htm>
- [20] Korea Electric Power Corporation. *Standard Supply Conditions for Specified Power Demand*. [Online]. Available: <http://www.kepco.co.kr>
- [21] Energy Agency of the Republic of Serbia. *Electricity Pricing for Tariff Customers*. [Online]. Available: <http://www.aers.rs/>
- [22] National Grid New York. *Understanding Electric Demand*. [Online]. Available: <https://www1.nationalgridus.com/CorporateHub>
- [23] Northwestern Public Service Company. *South Dakota Electric Rate Schedule No. 33 Commercial and Industrial Service*. [Online]. Available: <http://www.northwesternenergy.com>
- [24] Central Power Distribution Company of Andhra Pradesh Limited Annexure-A: *Retail Supply Tariff Schedule for FY 2013-14*. [Online]. Available: <http://www.apcentralpower.com>
- [25] Karnataka Electricity Regulatory Commission Bangalore. *Karnataka Electricity Regulatory Commission Code—Proposed Electric Power Tariff*. [Online]. Available: <http://www.kptcl.com/>
- [26] Virginia Electric and Power Company. *Schedule 6 Large General Service*. [Online]. Available: <https://www.dom.com/>
- [27] Uttar Gujarat VII Company Limited. *Tariff for Supply of Electricity at Low Tension, High Tension and Extra High Tension*. [Online]. Available: <http://www.ugvcl.com>
- [28] Singapore Power Group. *Use of System Charges*. [Online]. Available: <http://www.singaporepower.com.sg>
- [29] T. Hong, J. Wilson, and J. Xie, "Long term probabilistic load forecasting and normalization with hourly information," *IEEE Trans. Smart Grid*, vol. 5, no. 1, pp. 456–462, Jan. 2014.
- [30] R. Hyndman and S. Fan, "Density forecasting for long-term peak electricity demand," *IEEE Trans. Power Syst.*, vol. 25, no. 2, pp. 1142–1153, May 2010.
- [31] P. McSharry, S. Bouwman, and G. Bloemhof, "Probabilistic forecasts of the magnitude and timing of peak electricity demand," *IEEE Trans. Power Syst.*, vol. 20, no. 2, pp. 1166–1172, May 2005.
- [32] H. Morita, T. Kase, Y. Tamura, and S. Iwamoto, "Interval prediction of annual maximum demand using grey dynamic model," *Int. J. Elect. Power Energy Syst.*, vol. 18, no. 7, pp. 409–413, 1996.
- [33] A. Capasso, W. Grattieri, R. Lamedica, and A. Prudenzi, "A bottom-up approach to residential load modeling," *IEEE Trans. Power Syst.*, vol. 9, no. 2, pp. 957–964, May 1994.
- [34] E. Carpaneto and G. Chicco, "Probabilistic characterization of the aggregated residential load patterns," *IET Gener. Transm. Distrib.*, vol. 2, no. 3, pp. 373–382, May 2008.
- [35] J. Munkhammar, J. Rydén, and J. Widén, "Characterizing probability density distributions for household electricity load profiles from high-resolution electricity use data," *Appl. Energy*, vol. 135, pp. 382–390, Dec. 2014.
- [36] G. W. Irwin, W. Monteith, and W. C. Beattie, "Statistical electricity demand modelling from consumer billing data," *IEE Proc. C, Gener. Transm. Distrib.*, vol. 133, no. 6, pp. 328–335, Sep. 1986.
- [37] D. McQueen, P. Hyland, and S. Watson, "Monte Carlo simulation of residential electricity demand for forecasting maximum demand on distribution networks," *IEEE Trans. Power Syst.*, vol. 19, no. 3, pp. 1685–1689, Aug. 2004.
- [38] R. Herman and J. Kritzing, "The statistical description of grouped domestic electrical load currents," *Elect. Power Syst. Res.*, vol. 27, no. 1, pp. 43–48, 1993.
- [39] S. Heunis and R. Herman, "A probabilistic model for residential consumer loads," *IEEE Trans. Power Syst.*, vol. 17, no. 3, pp. 621–625, Aug. 2002.
- [40] R. Herman and S. Heunis, "Load models for mixed-class domestic and fixed, constant power loads for use in probabilistic LV feeder analysis," *Elect. Power Syst. Res.*, vol. 66, no. 2, pp. 149–153, 2003.



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